

$^{37}\text{Cl}(\text{p}, ^3\text{He})$ **1975Gu15**

$^{37}\text{Cl} \rightarrow ^{35}\text{S} + \text{p}1\text{n}$ from $J^\pi = 3/2^+$ ^{37}Cl ground state.

1975Gu15: A 40.2-MeV proton beam was produced from the Michigan State University cyclotron. The target was $55 \mu\text{g}/\text{cm}^2$ NaCl with 97% enriched ^{37}Cl made by evaporation of the salt onto a $30\text{--}\mu\text{g}/\text{cm}^2$ carbon backing. The reaction products were detected using a wire-counter plastic-scintillator in the focal plane of an Enge split-pole spectrograph with FWHM=30 keV. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced levels, L from zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. The target was natural purified chlorine gas 100 mm in diameter and 25 mm in height. The reaction particles were detected using two separate counter telescopes with each consisting of a $200\mu\text{m}$ phosphorous-drifted silicon ΔE detector, a 2-mm lithium-drifted silicon E detector and a 3-mm lithium-drifted silicon E-reject detector with FWHM=180 keV for ^3He . Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced L for ^{35}S ground state from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}S Levels

σ_{max} : From **1975Gu15**.

E(level) [†]	L [‡]	Comments
0	0+2+4	$\sigma_{\text{max}}=26.5 \mu\text{b}/\text{sr}$.
1575 10	0+2	$\sigma_{\text{max}}=19 \mu\text{b}/\text{sr}$.
1992 10	3+5	$\sigma_{\text{max}}=1.5 \mu\text{b}/\text{sr}$.
2717 10	0+2+4	$\sigma_{\text{max}}=53.9 \mu\text{b}/\text{sr}$.
2938 10	0+2+4	$\sigma_{\text{max}}=29.2 \mu\text{b}/\text{sr}$.
3421 10	0+2+4	$\sigma_{\text{max}}=81.5 \mu\text{b}/\text{sr}$.
3598 10	2+4	$\sigma_{\text{max}}=27.8 \mu\text{b}/\text{sr}$.
3811 10	3	$\sigma_{\text{max}}=2.6 \mu\text{b}/\text{sr}$.
4027 10	2	$\sigma_{\text{max}}=2.9 \mu\text{b}/\text{sr}$.
4114 10	0+2	$\sigma_{\text{max}}=8.8 \mu\text{b}/\text{sr}$.
4186 10	(2,3)	$\sigma_{\text{max}}=5.6 \mu\text{b}/\text{sr}$.
4290 10	(2)	$\sigma_{\text{max}}=2.8 \mu\text{b}/\text{sr}$.
4489 10	2	$\sigma_{\text{max}}=2.7 \mu\text{b}/\text{sr}$.
4577 10	0+2	$\sigma_{\text{max}}=10.7 \mu\text{b}/\text{sr}$.
4617 10	(1,2)	$\sigma_{\text{max}}=13.2 \mu\text{b}/\text{sr}$.
4843 10	2	$\sigma_{\text{max}}=15.9 \mu\text{b}/\text{sr}$.
4963 10	(0+2)	$\sigma_{\text{max}}=17.3 \mu\text{b}/\text{sr}$.
4990 10	0+2	$\sigma_{\text{max}}=19.5 \mu\text{b}/\text{sr}$.
5127 10	2	$\sigma_{\text{max}}=3.5 \mu\text{b}/\text{sr}$.
5345 10	3	$\sigma_{\text{max}}=2.2 \mu\text{b}/\text{sr}$.
5550 10	(3)	$\sigma_{\text{max}}=6.2 \mu\text{b}/\text{sr}$.
5771 10	2	$\sigma_{\text{max}}=10.1 \mu\text{b}/\text{sr}$.
5915? 10	(2,3)	$\sigma_{\text{max}}=3.4 \mu\text{b}/\text{sr}$.
6129 10	0+2	$\sigma_{\text{max}}=7.8 \mu\text{b}/\text{sr}$.
6347 10	(2)	$\sigma_{\text{max}}=4.8 \mu\text{b}/\text{sr}$.
6654 10	(3)	$\sigma_{\text{max}}=7.7 \mu\text{b}/\text{sr}$.
6696 10	2	$\sigma_{\text{max}}=6.7 \mu\text{b}/\text{sr}$.
7151? 10	(4)	$\sigma_{\text{max}}=3.3 \mu\text{b}/\text{sr}$.
7375? 10		
7712? 10	4	$\sigma_{\text{max}}=4.6 \mu\text{b}/\text{sr}$.
7770 10		$\sigma_{\text{max}}=5.5 \mu\text{b}/\text{sr}$.
8103 10	(1+3)	$\sigma_{\text{max}}=10.6 \mu\text{b}/\text{sr}$.
8160 10	(1)	$\sigma_{\text{max}}=8.4 \mu\text{b}/\text{sr}$.
8430 10	2	$\sigma_{\text{max}}=5.7 \mu\text{b}/\text{sr}$.
9155 10	2	$\sigma_{\text{max}}=10 \mu\text{b}/\text{sr}$.

Continued on next page (footnotes at end of table)

$^{37}\text{Cl}(\text{p}, ^3\text{He})$ [1975Gu15](#) (continued)

^{35}S Levels (continued)

[†] From [1975Gu15](#).

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in [1975Gu15](#). L is not considered firm by the evaluators when assigning J^π in the Adopted Levels because the theory of (p, ^3He) reactions is not robust.